

3.1 Characteristics of the Antarctic ecosystem

An **ecosystem** is a biological community of organisms that interact with each other and the physical environment. An ecosystem can range from a small wetland to an area that is thousands of square km in size, like the Great Barrier Reef off the coast of Australia. Organisms include animals, plants, and micro-organisms. These living organisms are called the **biotic** components of an ecosystem. Non-living things include the climate and soil. These physical features are called the **abiotic** components of an ecosystem.

The characteristics of an ecosystem, such as the amount and type of flora (plants) and the fauna (animals), depend on the abiotic factors found in an area. The **climate**, which includes factors such as rates of precipitation, amount of sunlight, temperature, and wind speed, all influence the types of plants that can grow in an area. Plants offer food and **habitat** for other organisms. The type of vegetation that can grow directly affects the fauna found in an area.

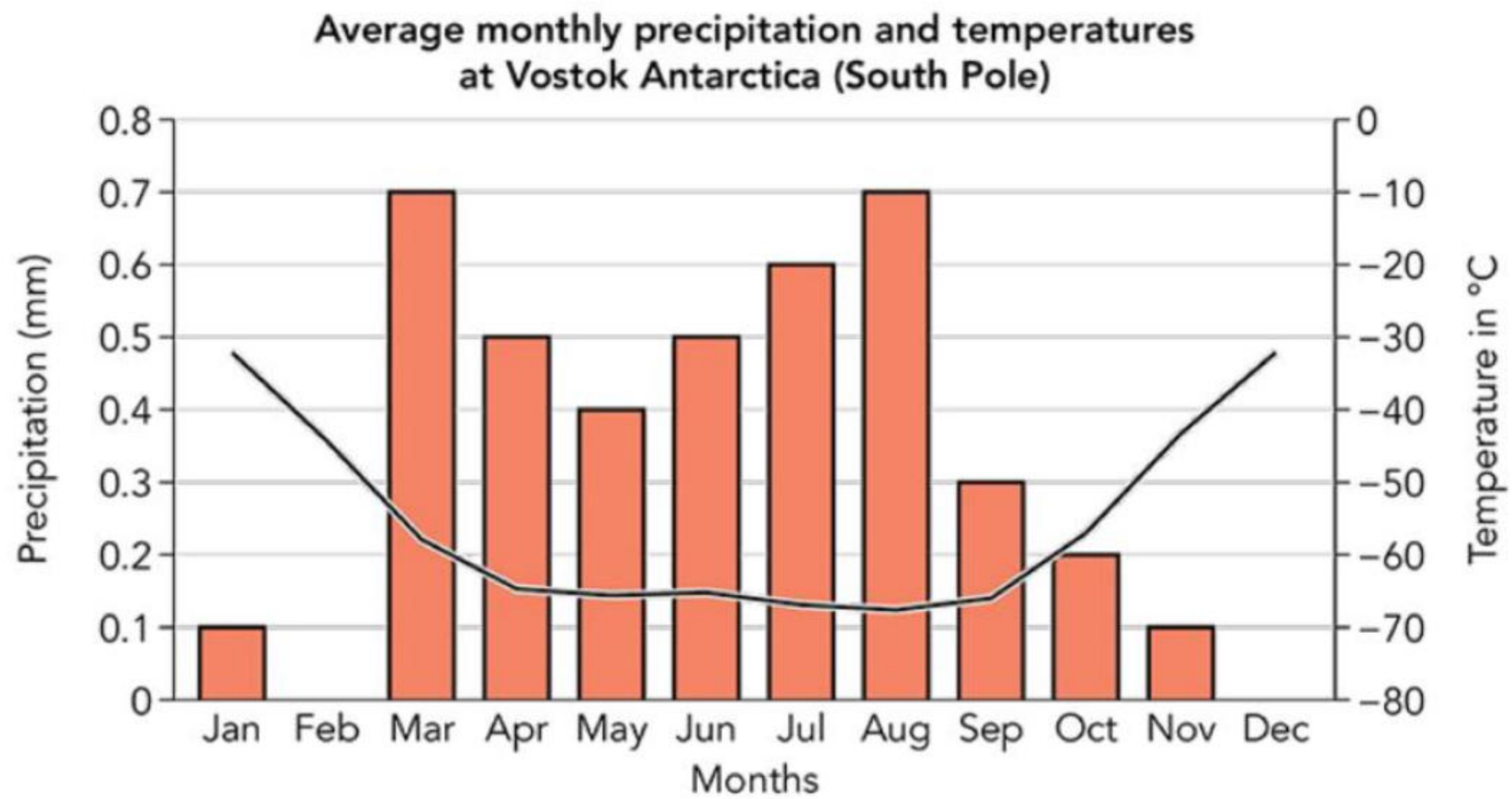
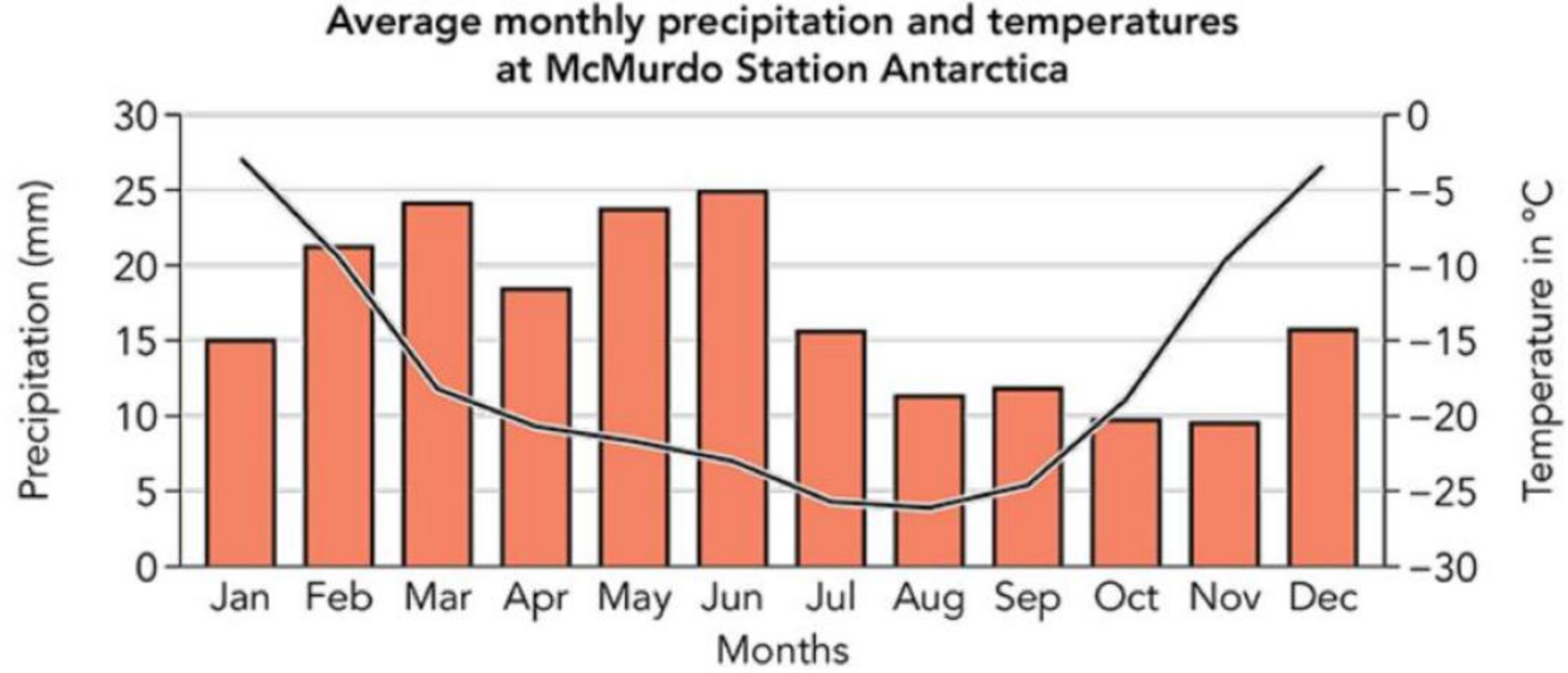
Location and features of the Antarctic ecosystem

Antarctica is the fifth-largest continent on Earth. It is located almost totally within the Antarctic circle and has ecosystems that have developed around the cold climate found in the region. It is entirely surrounded by the Southern Ocean and is the coldest place on Earth. The lowest temperature recorded was -93.2°C in 2010. Temperatures and precipitation vary widely across the continent, but the key characteristics of the climate are as outlined in Table 3.1.

Table 3.1: The climatic characteristics of the Antarctic ecosystems

Climate factor	Climatic factor range
Regional mean annual temperature	Mean annual temperature of the interior is -43.5°C Mean annual temperature of the coast is -10°C
Humidity	Often as low as 0.03%
Precipitation	Average for the continent is approximately 166 mm
Average seasonal temperature range	8°C to -60°C

Weather in Antarctica is highly variable, with conditions changing suddenly due to the strong winds that blow from the interior. The low rates of precipitation classify Antarctica as a cold desert (less than 250 mm of precipitation a year). Snowfalls rarely melt. Instead, snow forms compressed layers over time that create ice sheets. Nearly all of Antarctica is covered by the Antarctic ice sheet, which is at least 1500 metres thick.



Source **A**: Climate graphs for McMurdo and Vostok in Antarctica

► Image Description

TIP

When comparing two graphs, remember to read the scales. They may differ, and this will give you additional points to discuss.

TASK 1: Study Source **A**

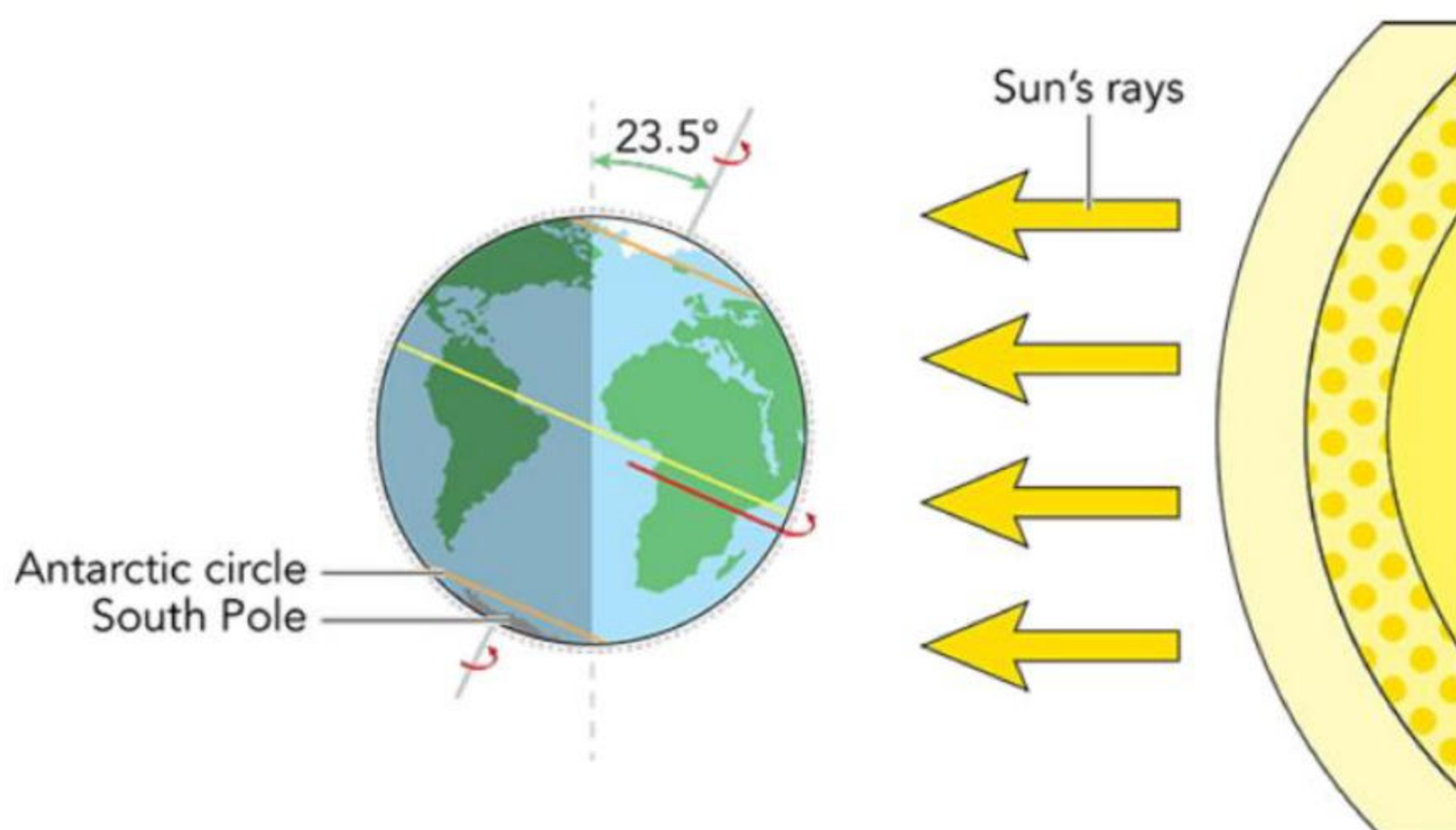
- Identify the coldest** temperature recorded in the two graphs.
- Which site has the lowest rate of precipitation?
- Calculate the difference** between the highest and lowest temperature for McMurdo.
- Compare the climate graph** for McMurdo Station with the one for Vostok. Use data to support your answer.

Factors influencing the characteristics of the Antarctic ecosystem

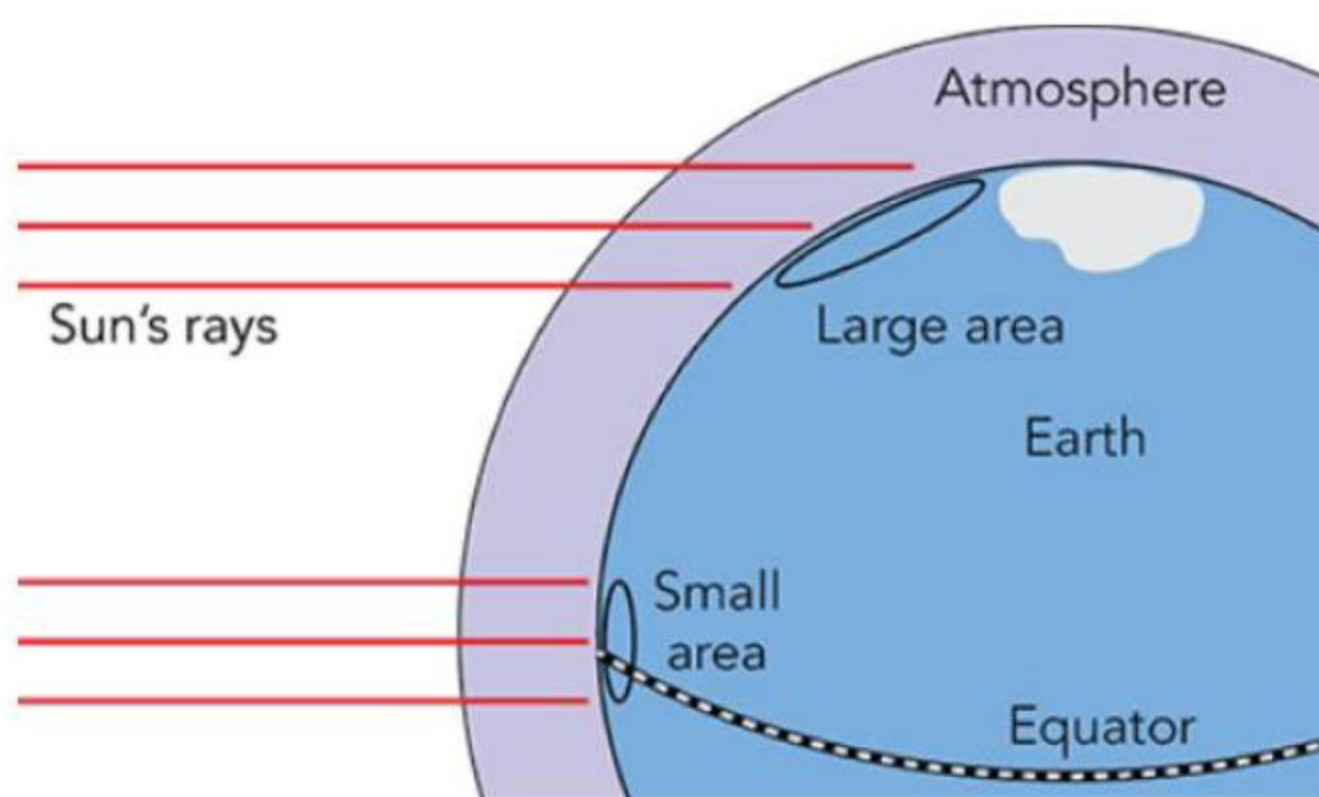
Several geographical factors cause the climate experienced in Antarctica. The Earth is tilted on the polar axis towards the sun at 23.5° (Source B), and the South Pole is located at a latitude of 90° south. This puts it at the furthest point from the equator. This tilt means that Antarctica's exposure to the sun changes with the Earth's orbit. This results in distinct seasons. Antarctica has 24 hours of daylight when the pole is facing towards the sun, and 24 hours of darkness when it is facing away from the sun.

The tilt of the Earth also means that any **insolation** in summer passes through the atmosphere and hits the Earth's surface at an oblique angle (Source C). The

sun's energy therefore loses strength and must warm a larger area of land than the same amount of insolation at the equator. This makes it less effective at heating up the Earth's surface, and results in much colder temperatures. These factors create a permanent high-pressure system over Antarctica (Source D). This results in dry cold air sinking over Antarctica's interior and limiting cloud formation and precipitation. In addition, the movement of air at the surface away from the high pressure towards the low pressure at the Antarctic circle (polar circle, 66.6°S) causes strong, unpredictable winds.

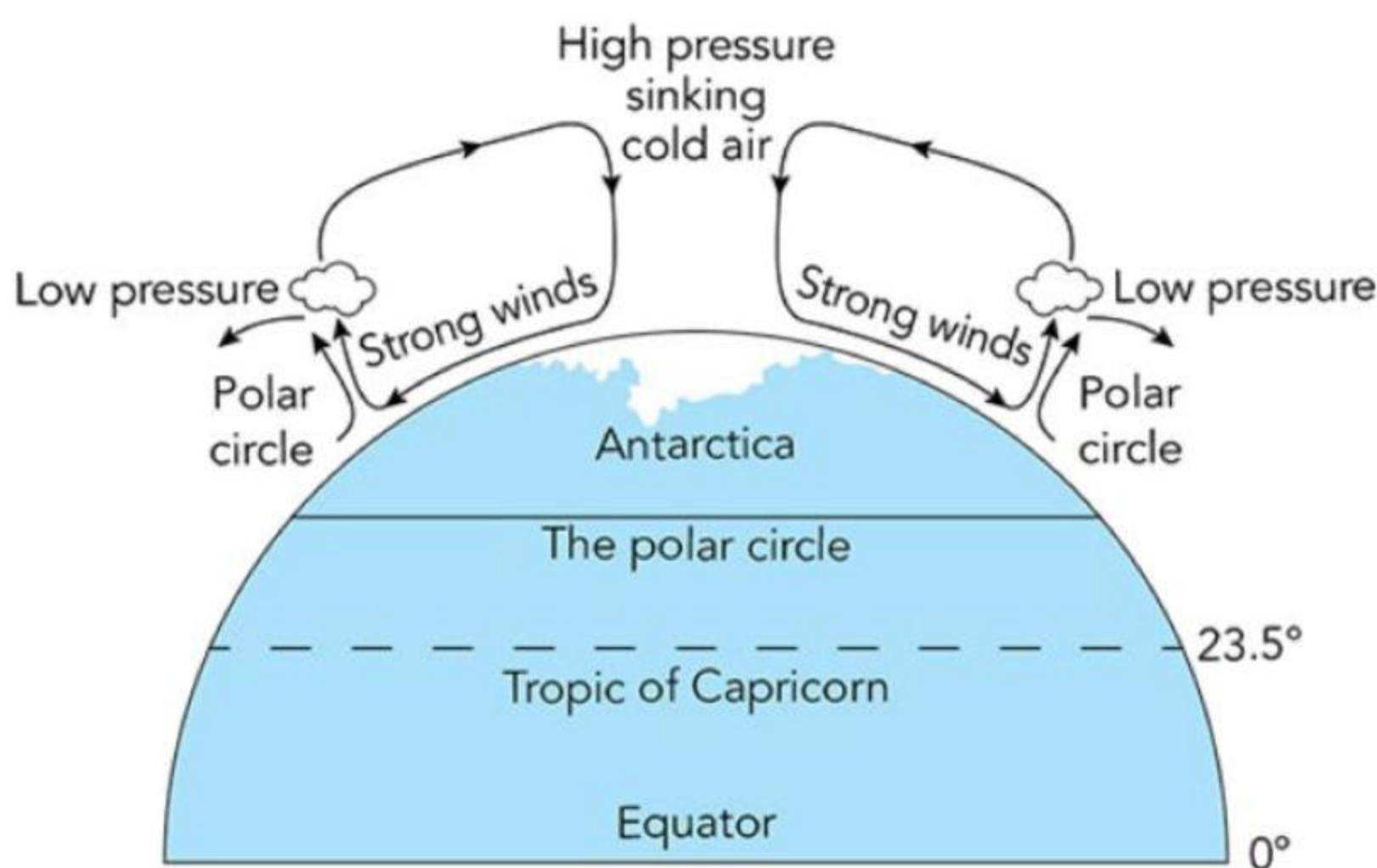


Source B: The position of the Earth in relation to the sun during the Antarctic winter



Source C: The rays from the sun travel a greater distance through the atmosphere and reach the poles at a more acute angle than the rays at the equator

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Source D: The permanent high-pressure system that brings dry, cold air to the inner continent of Antarctica and reduces the chances of precipitation

TASK 2: Study Sources B C D

a In small groups discuss how the following factors influence the climate in Antarctica:

- i The angle of the Earth towards the sun
 - ii The distance the sun's rays have to travel through the atmosphere
 - iii The area that the sun's rays must warm
 - iv The high-pressure atmospheric system
 - v Long dark winters (extreme seasonal influence)
- b Write a summary explaining why the climate of Antarctica is so extreme. Use information from the graphs in Source A to further develop your summary.

Abiotic and biotic factors of the Antarctic ecosystem

The abiotic factors (the extreme cold, lack of precipitation, lack of sunlight in winter, and weak sunlight in summer) influence the number and type of fauna and flora (the biotic factors) in Antarctica. Despite these factors, the marine ecosystem is rich in nutrients that are brought to the surface by ocean currents and support the complex food web.

Antarctica has both **terrestrial** and **marine ecosystems**. Terrestrial life in Antarctica is limited, without much of the plant life which commonly occurs on the other six continents. The vegetation on land is mainly moss, lichen, and algae. Springtails (terrestrial invertebrates) are one of the largest living terrestrial organisms in Antarctica, and only measure 3 mm in length. Organisms like penguins that live on both land and sea are not considered terrestrial.

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In contrast, the Antarctic marine ecosystem is complex, supported by microscopic phytoplankton that photosynthesises. The Antarctic marine food web supports the largest animal on Earth, the Antarctic blue whale, which can reach up to 29 metres in length and 199 tons in weight. The blue whale is a filter-feeder, relying on micro-organisms for survival.

Fauna in the Antarctic has adapted to the ecosystem by developing features such as:

- fat layers, which can provide insulation and energy storage (fat created in summer can provide energy during long winters with little food)
- feathers that repel water
- large eyes to spot prey in dark ocean waters
- behavioural adaptations, such as penguins huddling together to share warmth and conserve energy
- migratory patterns, seen in species that move north during winter to find food or reproduce.

Antarctica's flora consists mainly of algae, lichens, moss, fungi, and Antarctic hair grass and pearlwort. They are all able to photosynthesise at very low temperatures and have adapted in various ways. For example, the hair grass and pearlwort can survive trampling of seals and penguin colonies. These species also flower and produce seeds in the long daylight hours of summer, relying on pollination by wind due to the lack of insects. The algae, lichen, moss, and fungi lack root systems, so don't require a complex intake of nutrients. They are also able to absorb water easily.



Source **E**: Penguins diving into the cold ocean off an ice shelf in Antarctica

REFLECTION

How did you choose which information to include on the poster? Did you find it hard agreeing with your partner? Did the process of comparing ideas and coming to an agreement help you make a better choice? How could you improve on working in pairs in the future?

TASK 3: Study Source **E**

- a** In pairs, discuss how the environment in Antarctica has protected the species that live there.
- b** With a partner, carry out research on one Antarctic animal or organism and identify how it has adapted to survive in the cold climate. Create a spider diagram on a small poster giving information on the adaptations of the species you have researched.